

Density functional theory for sustainable energy materials

CENSEMAT DFT-ML Workshop. Aarhus University.

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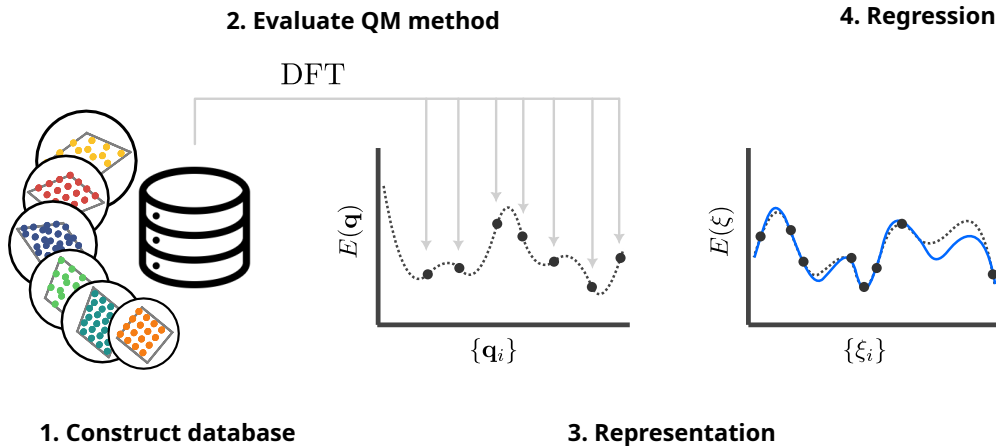
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04.05.26



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Machine-learning interatomic potentials (MLIPs)



The electronic Schrödinger equation

$$\hat{\mathcal{H}}_e \Psi_n(\mathbf{r}_{1-N} \sigma_{1-N}) = E_n \Psi_n(\mathbf{r}_{1-N} \sigma_{1-N})$$

Nuclear positions enter the electronic Hamiltonian as parameters:

$$\hat{\mathcal{H}}_e = \underbrace{-\sum_i \frac{\hat{V}_i^2}{2}}_{\hat{T}_e} + \underbrace{\sum_i \sum_A \frac{\overbrace{-Z_A}^{V_{\text{ext}}(\mathbf{r}_i)}}{|\hat{\mathbf{r}}_i - \mathbf{R}_A|}}_{\hat{V}_{eN}} + \underbrace{\frac{1}{2} \sum_{j \neq i} \frac{1}{|\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|}}_{\hat{V}_{ee}}$$

Represent wave function on 20 points for each component

Helium $\Psi(\mathbf{r}_1, \mathbf{r}_2)$
 $20^{(3 \cdot 2)} \approx 64 \cdot 10^6$ values needed

Titanium, $\Psi(\{\mathbf{r}_{1-22}\})$
 $20^{(3 \cdot 22)} \approx 7 \cdot 10^{85}$ values needed

Wave function depends simultaneously on the coordinates of all electrons

The variational principle

Energy functional

$$\begin{aligned} E[\Psi] &= \int \Psi^* \hat{T}_e \Psi d\mathbf{r}_{1-N} d\sigma_{1-N} + \int \Psi^* \hat{V}_{eN} \Psi d\mathbf{r}_{1-N} d\sigma_{1-N} + \int \Psi^* \hat{V}_{ee} \Psi d\mathbf{r}_{1-N} d\sigma_{1-N} \\ &= \langle \Psi | \hat{T}_e | \Psi \rangle + \langle \Psi | \hat{V}_{eN} | \Psi \rangle + \langle \Psi | \hat{V}_{ee} | \Psi \rangle \\ &= E_{kin}[\Psi] + E_{ext}[\Psi] + E_{ee}[\Psi] \end{aligned}$$

The electronic ground-state wave function

Lowest energy of all normalized antisymmetric wave functions

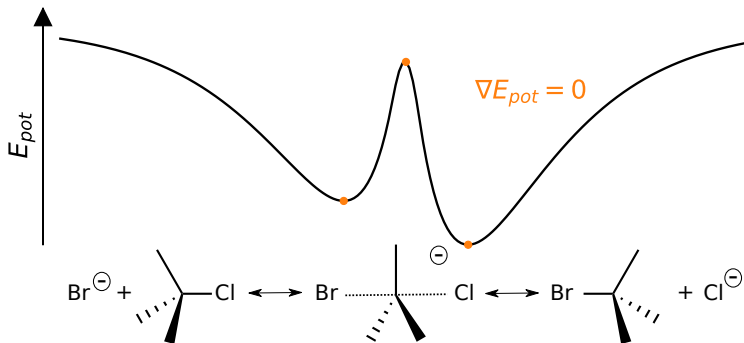
Constrained search:

$$E_0 = \min_{\langle \Psi | \Psi \rangle = 1} (E[\Psi]) = E[\Psi_0]$$

Potential energy surface

Parametric dependence of electronic energy on the nuclear positions

$$E_{pot}(\{\mathbf{R}_A\}) = E[\psi_0] + \frac{1}{2} \sum_{A_1 \neq A_2} \frac{Z_{A_1} Z_{A_2}}{|\mathbf{R}_{A_1} - \mathbf{R}_{A_2}|}$$



Density functionals for indistinguishable particles

External energy

$$E_{\text{ext}}[n] = \int n(\mathbf{r}) V_{\text{ext}}(\mathbf{r}) d\mathbf{r}$$

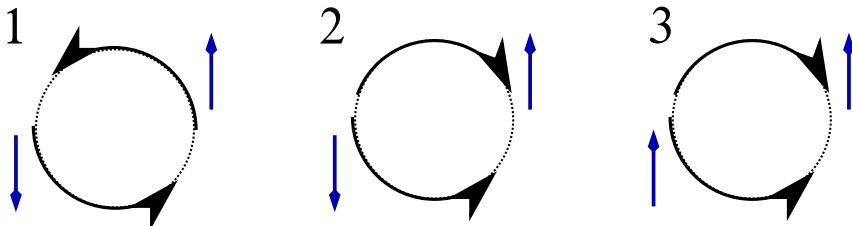
Electron-electron repulsion

$$\begin{aligned} E_{ee}[\Pi] &= \frac{1}{2} \int \frac{\Pi(\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2)}{r_{12}} d\mathbf{r}_1 d\mathbf{r}_2 d\sigma_1 d\sigma_2 \\ &= \underbrace{\frac{1}{2} \int \frac{n(\mathbf{r}_1)n(\mathbf{r}_2)}{r_{12}} d\mathbf{r}_1 d\mathbf{r}_2}_{E_H} + \underbrace{\frac{1}{2} \int \frac{\Pi_{xc}(\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2)}{r_{12}} d\mathbf{r}_1 d\mathbf{r}_2 d\sigma_1 d\sigma_2}_{U_{xc}} \end{aligned}$$

Hartree energy exchange-correlation energy

Correlation

Three classic situations with the same probability density but different pair probability densities:



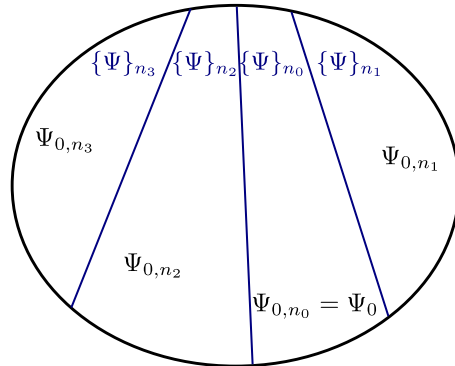
1. The particles never meet
2. The particles with anti-parallel spin meet with a double rotation frequency
3. The particles with parallel spin meet with a double rotation frequency

DFT. Constrained search formulation.

1. Take all possible antisymmetric wave functions for N electrons
2. Divide them into groups such that all wave functions in a group result in the same density
3. Define energy functional

$$F_{HK}[n] = \min_{\{\Psi\}_n} (E_{kin}[\Psi] + E_{ee}[\Psi])$$

$$E[n] = F_{HK}[n] + \int n(\mathbf{r})V_{ext}(\mathbf{r})d\mathbf{r}$$



Minimize the energy for a given external potential with respect to the trial densities

$$E_0 = \min_{n(\mathbf{r})} (F_{HK}[n(\mathbf{r})] + \int n(\mathbf{r})V_{ext}(\mathbf{r})d\mathbf{r}) = E[n_0]$$

Energy functionals

Thomas-Fermi

$$E_{TF}[n] = \int \tau^{HEG}[n] d\mathbf{r} + E_{ext}[n] + E_H[n] + \int e_{xc}^{HEG}[n] d\mathbf{r}$$

Slater determinant

$$E[\{\psi_i\}] = T_s[\{\psi_i\}] + E_{ext}[n] + E_H[n] + E_x[\{\psi_i\}]$$

$$n(\mathbf{r}) = \sum_i \psi_i^*(\mathbf{r}) \psi_i(\mathbf{r}) \quad , \quad T_s = \sum_i \langle \psi_i | -\frac{\nabla^2}{2} | \psi_i \rangle$$

The Kohn-Sham energy functional



"I knew that the Hartree equations described atomic ground states much better than TF theory. The difference between them lay in the different treatments of the kinetic energy"

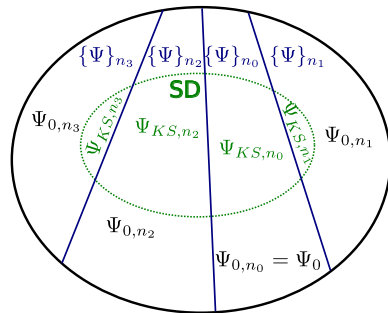
Kohn *Rev. Mod. Phys.* **71**, p1253

$$E_{KS}[n] = T_s[n] + E_{ext}[n] + E_H[n] + E_{xc}[n]$$

The Kohn-Sham kinetic energy functional

Introduce auxiliary system of non-interacting electrons with SD wavefunction

$$T_s[n] = \min_{\{\Psi_{SD}\}_n} \left(\left\langle \Psi \left| \sum_i -\frac{\nabla^2}{2} \right| \Psi \right\rangle \right) \\ = \left\langle \Psi_{KS,n} \left| \sum_i -\frac{\nabla^2}{2} \right| \Psi_{KS,n} \right\rangle$$



The minimal kinetic energy of non-interacting particles that reproduce a given density

$$T_s[n] = \min_n(\mathbf{r}) \left(\sum_i \left\langle \psi_i \left| -\frac{\nabla^2}{2} \right| \psi_i \right\rangle \right) , \quad n(\mathbf{r}) = \sum_i \psi_i^*(\mathbf{r}) \psi_i(\mathbf{r})$$

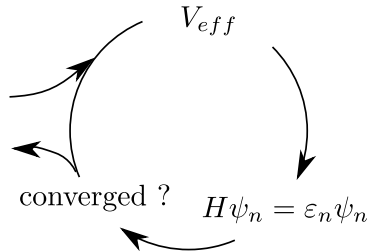
The Kohn-Sham equations

Minimise E_{KS} under the constraint of orthogonal orbitals

$$\begin{aligned} V_{eff}(\mathbf{r}) &= \frac{\partial E_{ext}}{\partial n(\mathbf{r})} + \frac{\partial E_H}{\partial n(\mathbf{r})} + \frac{\partial E_{xc}}{\partial n(\mathbf{r})} \\ &= V_{ext}(\mathbf{r}) + V_H(\mathbf{r}) + V_{xc}(\mathbf{r}) \end{aligned}$$

$$\left(-\frac{\nabla^2}{2} + V_{eff}(\mathbf{r}) \right) \psi_n(\mathbf{r}) = \epsilon_n \psi_n(\mathbf{r})$$

One-electron Schrödinger-like equations describing non-interacting electrons in an effective potential, which is such that, if E_{xc} is exact, the resulting density is the exact ground state density.



Questions

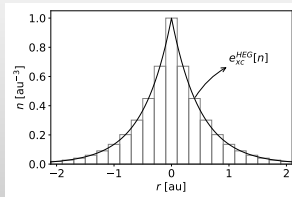
- Why can the ground-state energy be written as a functional of the density when the correlation energy can not ?
- What are the ideas behind the KS energy ?



The local density approximation

The exchange-correlation energy density, e_{xc} , at each point is approximated by that of a homogeneous electron gas with the same density.

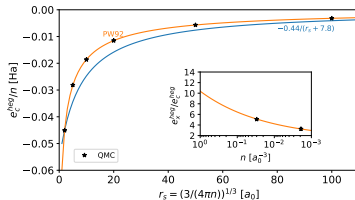
$$E_{xc}^{LDA}[n] = \int e_{xc}^{HEG}[n(\mathbf{r})] d\mathbf{r} = \int (e_x^{HEG}[n(\mathbf{r})] + e_c^{HEG}[n(\mathbf{r})]) d\mathbf{r}$$



Exchange

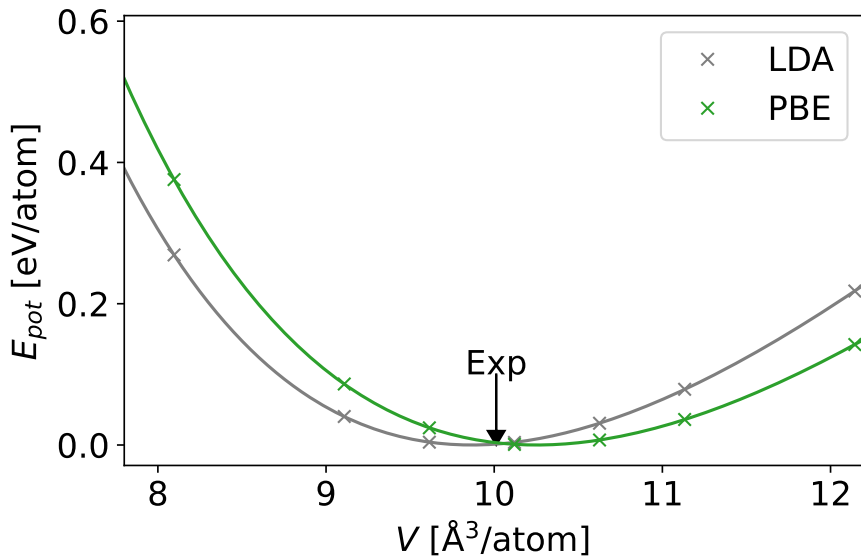
$$e_x^{HEG} = -\frac{3}{4} \left(\frac{3}{\pi} \right)^{1/3} n^{4/3}$$

Correlation



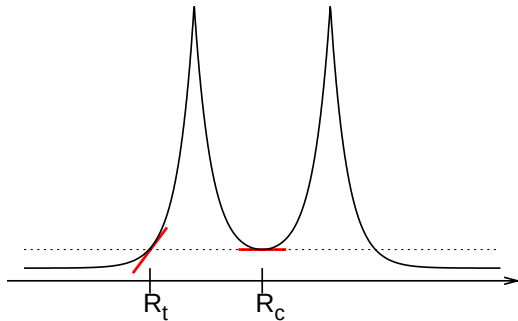
Kohn-Sham (1965): “We do not expect an accurate description of chemical binding.”

Si volume optimization



The generalized gradient approximation

$$E_{xc}^{GGA} = \int e_{xc}[n, |\nabla n|] d\mathbf{r} = \int e_{xc}^{LDA}[n] F_{xc}^{GGA}[n, |\nabla n|] d\mathbf{r}$$



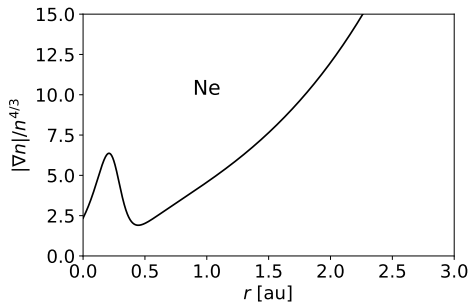
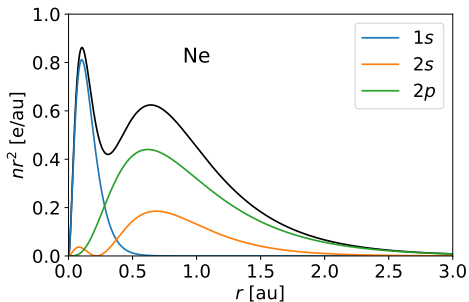
Inclusion of the gradient allows to discriminate between points with similar densities but different (local) surroundings

Lowest gradient correction

$$e_x[n, \nabla n] = e_x^{LDA}[n](1 + ax^2) \quad , \quad x = \underbrace{\frac{|\nabla n|}{n^{4/3}}}_{\text{reduced gradient}}$$

The reduced gradient diverges for exponential decreasing densities

$$n(r) = A \exp(-ar) \quad , \quad \frac{|\nabla n|}{n^{4/3}} = \frac{an}{n^{4/3}} = an^{-1/3}$$



PBE exchange

GGA should retain conditions satisfied by LDA and use flexibility to add others

$$e_x^{PBE}(n, |\nabla n|) = e_x^{HEG}(n) F_x^{PBE}(s) \quad , \quad F_x^{PBE}(s) = 1 + \frac{\mu s^2}{1 + \mu s^2 / \kappa} \quad , \quad s = \frac{1}{2(3\pi^2)^{1/3}} \frac{|\nabla n|}{n^{4/3}}$$

Slowly varying limit

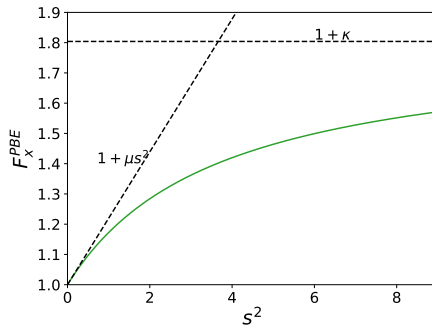
$$F_x(s^2) \rightarrow 1 + \mu s^2 \quad , \quad s^2 \rightarrow 0$$

x and c gradient corrections cancel

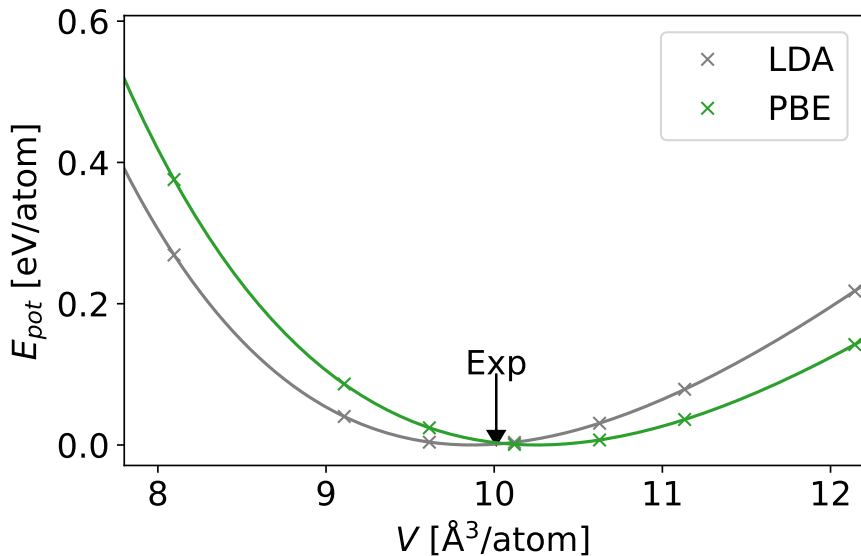
Local upper bound

$$F_x \rightarrow 1 + \kappa \quad \rightarrow \quad s^2 \rightarrow \infty$$

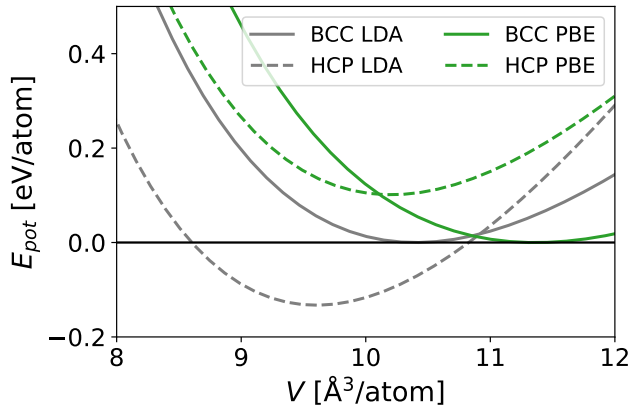
“Global Lieb-Oxford” upper bound for E_x



Si volume optimization



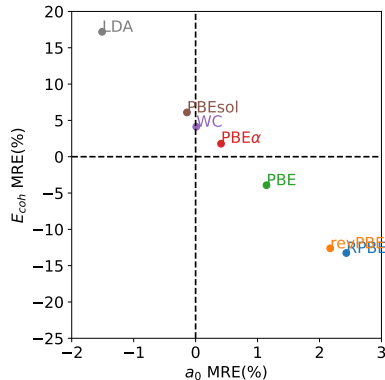
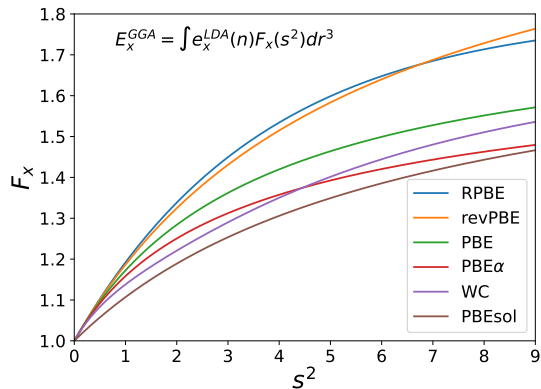
Stabilization of magnetism in Fe



	LDA	PBE
$m(a_{exp}^{BCC}) (\mu_B)$	2.21	2.22

GGAs stabilize open structures, which in turn stabilize magnetism

The GGA functional ZOO



- “Small” enhancement: Good for lattice constants
- “Large” enhancement: Good for cohesive energies

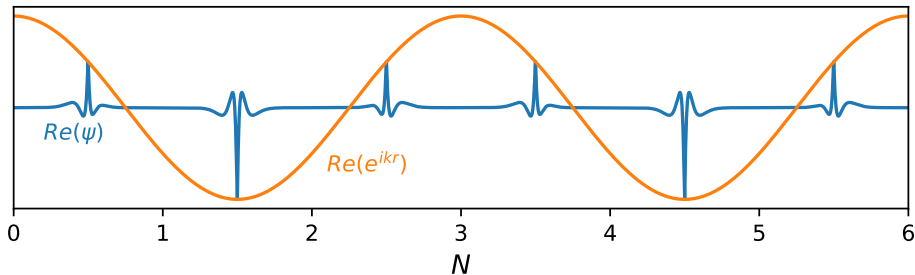
- What is the local density approximation ?
- How do GGAs improve on the accuracy the LDA ?
- Why do we not have only one GGA ?

Bloch waves

Translational symmetry: Norm of one-electron wave functions must be periodic

$$\psi_{n\mathbf{k}}(\mathbf{r}) = u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\mathbf{r}} \quad , \quad u_{n\mathbf{k}}(\mathbf{r}) = u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}_l)$$

Equivalent formulation $\psi_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}_l) = \psi_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\mathbf{R}_l}$

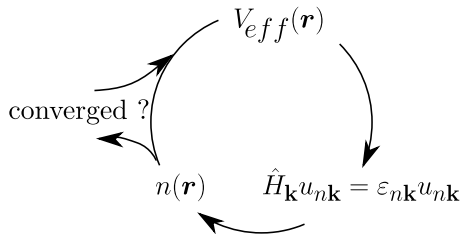


The Kohn-Sham equations using Bloch waves

$$\nabla^2 \psi_{n\mathbf{k}} = (\nabla + i\mathbf{k})^2 u_{n\mathbf{k}} e^{i\mathbf{k}\mathbf{r}}$$

Decoupled Kohn-Sham equations for periodic part of each Bloch state

$$\underbrace{\left(-\frac{(\nabla + i\mathbf{k})^2}{2} + V_{eff} \right)}_{\hat{H}_{\mathbf{k}}} u_{n\mathbf{k}} = \varepsilon_{n\mathbf{k}} u_{n\mathbf{k}}$$



$$n(\mathbf{r}) = \sum_{n\mathbf{k}} w_{n\mathbf{k}} |u_{n\mathbf{k}}(\mathbf{r})|^2$$

“AO” basis

Bloch: $\psi_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = \psi_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\mathbf{R}}$

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mu\mathbf{l}} c_{\mu}^{(n\mathbf{k})} e^{i\mathbf{k}\mathbf{R}_{\mathbf{l}}} \phi_{\mu\mathbf{l}}(\mathbf{r})$$

Efficient but difficult to improve systematically

Plane-wave basis

Bloch: $\psi_{n\mathbf{k}}(\mathbf{r}) = u_{n\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\mathbf{r}}$

$$u_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{l}} c_{\mathbf{K}_{\mathbf{l}}}^{(n\mathbf{k})} e^{i\mathbf{K}_{\mathbf{l}}\mathbf{r}} \quad , \quad \psi_{n\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{l}} c_{\mathbf{K}_{\mathbf{l}}}^{(n\mathbf{k})} e^{i(\mathbf{K}_{\mathbf{l}}+\mathbf{k})\mathbf{r}}$$

Straightforward systematic improvement but inefficient

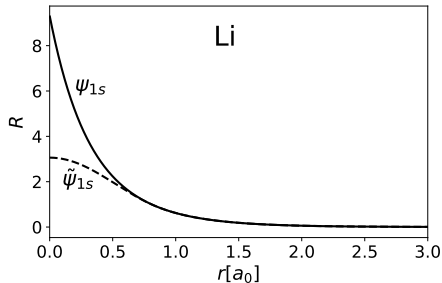
Projector augmented waves

Linear transformation

$$|\psi_n\rangle = \hat{T} |\tilde{\psi}_n\rangle \quad , \quad \hat{T} = 1 + \sum_A \hat{T}_A$$

$|\psi_n\rangle$ Kohn-Sham orbitals

$|\tilde{\psi}_n\rangle$ Pseudo wave functions: Numerically convenient (a.k.a. smooth) auxiliary functions



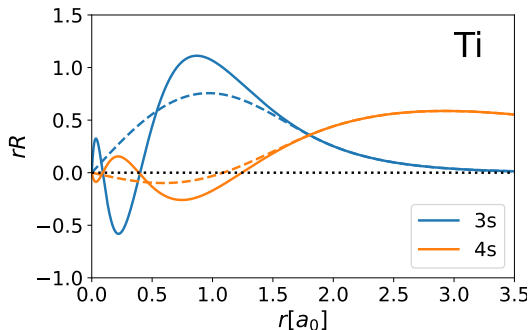
evaluate the total energy directly as a functional of the pseudo wave functions.

PAW: Partial waves

Modify auxiliary functions in regions around each nucleus ($i = j_l, m$).

$$\hat{T} |\tilde{\psi}_n\rangle = \left(1 + \sum_a \hat{T}^a\right) |\tilde{\psi}_n\rangle = |\tilde{\psi}_n\rangle + \sum_a \sum_{i \in a} (|\phi_i^a\rangle - |\tilde{\phi}_i^a\rangle) \langle \tilde{p}_i^a | \tilde{\psi}_n \rangle$$

- $|\phi_i^a\rangle$: Atomic waves
- $|\tilde{\phi}_i^a\rangle$: Smooth functions with $\tilde{\phi}_i^a(r) = \phi_i^a(r)$ for $r > r_{a,i}$



Atomic corrections

$$O = \underbrace{\sum_n f_n \langle \tilde{\psi}_n | \hat{O} | \tilde{\psi}_n \rangle}_{\text{Auxiliary contribution}} + \underbrace{\sum_a (O_c^a + \sum_{i_1 i_2} D_{i_1 i_2}^a \Delta O_{i_1 i_2}^a)}_{\text{Atomic corrections}}$$

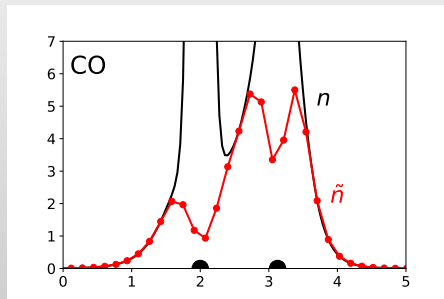
$$\begin{aligned} D_{i_1 i_2}^a &= \sum f_n \langle \tilde{\psi}_n | \tilde{p}_{i_1}^a \rangle \langle \tilde{p}_{i_2}^a | \tilde{\psi}_n \rangle \\ \Delta O_{i_1 i_2}^a &= \langle \phi_{i_1}^a | \hat{O} | \phi_{i_2}^a \rangle - \langle \tilde{\phi}_{i_1}^a | \hat{O} | \tilde{\phi}_{i_2}^a \rangle \\ O_c^a &= \sum \langle \phi_c^a | \hat{O} | \phi_c^a \rangle \end{aligned}$$

Kinetic energy

$$\begin{aligned} T_s &= \sum_n f_n \langle \tilde{\psi}_n | -\frac{\nabla^2}{2} | \tilde{\psi}_n \rangle + \sum_a (T_c^a + \sum_{i_1 i_2} D_{i_1 i_2}^a \Delta T_{i_1 i_2}^a) \\ T_s &= \tilde{T}_s[\{\tilde{\psi}_n\}] + \sum_a (T_c^a + \Delta T^a[\{\tilde{\psi}_n\}]) \end{aligned}$$

$$\hat{O} = \delta(\mathbf{r} - \mathbf{r}_1) : \quad n(\mathbf{r}) = \sum_n f_n \tilde{\psi}_n^*(\mathbf{r}) \tilde{\psi}_n(\mathbf{r}) + \sum_a \left(n_c^a(\mathbf{r}) + \sum_{i_1 i_2} D_{i_1 i_2}^a (\phi_{i_1}^a(\mathbf{r}) \phi_{i_2}^a(\mathbf{r}) - \tilde{\phi}_{i_1}^a(\mathbf{r}) \tilde{\phi}_{i_2}^a(\mathbf{r})) \right)$$

$$n(\mathbf{r}) = \tilde{n}(\mathbf{r}) + \sum_a n_c^a(\mathbf{r}) + \Delta n^a(\mathbf{r})$$



- PAW formalism
- Dual atomic and grid/plane-wave basis
- Highly parallelized
- Python based
- GNU Public License

